# High-Yield ENT Reviewer

Head & Neck Surgery — board / in-service & consultant "pimp" review

Built from Catherine's notebook and reorganized into a rapid reviewer. Primary framing: **Cummings Otolaryngology**–consistent teaching, cross-checked against current guidelines (ACR TI-RADS, Bethesda 2017, AAO-HNS). **Highlighted** = highest-yield. Not a substitute for the textbook; confirm edition-specific numbers against your copy of Cummings.

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## 1. Facial & Palate Embryology

| Week | Event |
|---|---|
| 4 | Five facial prominences appear: 1 **frontonasal**, paired **maxillary**, paired **mandibular**. |
| 5 | Nasal placodes → **nasal pits**, splitting frontonasal into **medial** & **lateral nasal prominences**. |
| 6–7 | Maxillary prominences fuse with medial nasal prominences → **upper lip**; merged medial nasal prominences → intermaxillary segment. |
| ~10 | **Lip** fusion complete. |
| 7–12 | **Palate** fuses **anterior → posterior**, ending at the **uvula (~wk 12)**. |

| Prominence | Gives rise to |
|---|---|
| **Medial nasal** (merged) | Philtrum, nasal tip, columella, **primary palate (premaxilla)**, 4 incisors |
| **Maxillary** | Lateral upper lip, cheeks, **secondary palate** |
| **Lateral nasal** | **Nasal alae** (sides) |
| **Mandibular** | Lower lip, mandible |

**PEARL** The **incisive foramen** divides **primary palate (anterior)** from **secondary palate (posterior)** — the key landmark for classifying clefts. Primary = fused medial nasal prominences; secondary = fused palatal shelves of the maxillary prominences.

**PIMP** "What structures form the philtrum?" → the merged **medial nasal prominences**. "Why does an isolated soft-palate cleft exist but not an isolated hard-palate cleft?" → palate closes **front to back**, so the soft palate is last to fuse.

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## 2. Cleft Lip & Palate

### Epidemiology & associations

- **Cleft lip ± palate:** more common in **males**; **left-sided** is the most common unilateral site. Isolated **cleft palate** is more common in **females**.
- Non-syndromic mostly; **Pierre Robin sequence** = micrognathia → glossoptosis → U-shaped cleft palate + airway obstruction (tongue mechanically blocks shelf fusion; the note's "macroglossia" is a slip — tongue is normal-sized but retropositioned).
- **Van der Woude syndrome** = most common syndromic cause of CL/P (lower-lip pits).

### Teratogens / risk factors

- **Low folic acid** (supplementation protective), **anti-epileptics** (phenytoin, valproate), **retinoids/ isotretinoin**, maternal smoking/alcohol.
- Prenatal ultrasound detects clefts ~**18–20 wks** (anomaly scan).

### Submucous cleft palate — Calnan's triad

- **Bifid uvula** + **zona pellucida** (bluish midline, deficient musculus uvulae) + **notch in posterior hard palate**.
- Abnormal levator/tensor sling → Eustachian tube dysfunction → **recurrent OME / conductive hearing loss**.

### Surgical timeline & the "Rule of 10s"

| Procedure | Timing | Key point |
|---|---|---|
| **Cheiloplasty** (lip) | ~3 mo (**Rule of 10:** 10 wks, 10 lb, Hgb 10) | Earlier → maxillary retrusion |
| **Palatoplasty** | **~9–18 mo (before speech)** | Delay → compensatory speech / VPI |
| **Alveolar bone graft** | ~7–9 y (mixed dentition) | Timed to canine eruption (root ⅓–⅔ formed) |
| Definitive rhinoplasty | ~14 y (F) / 16 y (M) | After skeletal maturity |

### Palatoplasty techniques

| Technique | Principle | Note |
|---|---|---|
| **Von Langenbeck** | Bipedicle mucoperiosteal flaps + lateral relaxing incisions | Oldest; good blood supply, limited lengthening |
| **Veau-Wardill-Kilner** | V-Y **push-back** (greater palatine pedicle) | Lengthens velum; raw area anteriorly |
| **Furlow** | **Double-opposing Z-plasty** | Realigns levator sling → best speech & lowest OME |

**PIMP** "Which palatoplasty best restores the levator sling / gives the best speech?" → **Furlow double-opposing Z-plasty.** "Why repair the palate by ~1 year?" → to precede meaningful speech development and avoid velopharyngeal insufficiency.

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## 3. Sinonasal Anatomy & Epistaxis

### Paranasal sinus development (pneumatization)

- **Maxillary & ethmoid** — present at birth.
- **Sphenoid** — from ~3 y. **Frontal** — last; ~7–10 y, adult size in adolescence.

### Sinus drainage

| Drains to | Sinuses |
|---|---|
| **Middle meatus** (via OMC) | Anterior ethmoid, **maxillary**, frontal |
| Superior meatus | Posterior ethmoid |
| Sphenoethmoidal recess | Sphenoid |
| Inferior meatus | Nasolacrimal duct (via **valve of Hasner**) |

### Blood supply / epistaxis

- **Kiesselbach's plexus (Little's area)**, anterior septum = anastomosis of **anterior ethmoidal, sphenopalatine, greater palatine, superior labial** arteries → most **anterior epistaxis** (most common overall, esp. children).
- **Woodruff's plexus** (posterolateral, under inferior turbinate) → **posterior epistaxis** (older, hypertensive).
- Sphenopalatine artery = terminal branch of the **internal maxillary artery** (from external carotid). Anterior/posterior ethmoidals come from the **ophthalmic** → **internal carotid**.

**PEARL** **Best initial management of epistaxis:** firm compression of the alae + topical vasoconstrictor; then anterior packing. **Refractory posterior bleed:** SPA ligation/cautery or embolization. Ligating the ethmoidals will not stop an ICA-territory bleed control from the ECA side.

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## 4. Chronic Rhinosinusitis (CRS) & Nasal Polyps

### Diagnosis (need symptoms ≥12 wks + objective evidence)

- **≥2 symptoms:** nasal obstruction, mucopurulent discharge, facial pain/pressure, hyposmia — **≥12 weeks**.
- **Objective:** **nasal endoscopy** (polyps, edema, purulence) **or** CT.
- **Gold-standard imaging = non-contrast CT of paranasal sinuses (PNS).**

### Lund-Mackay CT staging

- Each sinus system: **0 clear** / **1 partial** / **2 complete** opacification; each OMC: **0 or 2.**
- **Max 12 per side → 24 total.**

### Polyp scores (don't confuse them)

| System | What it grades |
|---|---|
| **Endoscopic polyp grade (0–3)** | Extent: 0 none → 1 within middle meatus → 2 below middle turbinate → 3 beyond |
| **Lund-Kennedy** | Endoscopic appearance: polyp, edema, discharge, scarring, crusting |
| **Lund-Mackay** | CT opacification (above) |

### Medical management (CRS)

- **Intranasal corticosteroids** (mainstay); short oral steroid course for polyps / pre-op field.
- **High-volume, low-pressure saline irrigation.**
- **Long-term low-dose macrolide** (e.g., clarithromycin up to ~12 wks) as anti-inflammatory **immunomodulator** in CRSsNP.
- CRSwNP: consider **biologics** (anti–IL-4/5/IgE) in refractory disease.

### Special polyps / differentials

- **Antrochoanal polyp (Killian polyp):** solitary, from **maxillary sinus → middle meatus → choana**; young patients; **unilateral** — treat surgically (recurs if origin not removed).
- **Samter's triad** = asthma + nasal polyps + **aspirin/NSAID sensitivity** (AERD).

**RED FLAG** **Unilateral** nasal polyp/mass in a child/young male → think **antrochoanal polyp or JNA** (biopsy JNA is dangerous — image first). Unilateral mass in an adult → rule out **inverted papilloma** / **malignancy**.

## 5. FESS & CT Anatomy

### Goals of FESS

- Restore **mucociliary clearance** — widen the OMC, re-establish natural drainage, improve topical drug delivery. Correct accessory-ostium **recirculation**.

### Ethmoid lamellae (anterior → posterior)

| # | Lamella | Note |
|---|---|---|
| 1 | **Uncinate process** | Anterior border of hiatus semilunaris |
| 2 | Ethmoid bulla | Largest anterior ethmoid cell |
| 3 | **Basal (ground) lamella of middle turbinate** | Divides **anterior vs posterior ethmoid** |
| 4 | Superior turbinate | Leads to sphenoethmoidal recess |

### High-yield ethmoid cells

| Cell | Location / danger |
|---|---|
| **Agger nasi** | Most anterior ethmoid cell (frontal recess) |
| **Haller cell** | Infraorbital; narrows maxillary ostium |
| **Onodi cell** | Posterior ethmoid near **optic nerve & ICA** — surgical danger |

### FESS complications

| Major | Minor |
|---|---|
| Orbital hematoma, **diplopia** (medial rectus via lamina papyracea), **CSF leak (cribriform/skull base)**, ICA injury, major hemorrhage, blindness | Epistaxis, synechiae/ adhesions |

**PIMP** "Most medial ethmoid wall you can injure → orbit?" **Lamina papyracea.** "Which cell risks the optic nerve?" **Onodi.** "Roof of the ethmoid / thinnest point for CSF leak?" **Cribriform plate (lateral lamella).**

## 6. Orbital Complications of Sinusitis — Chandler

| Stage | Entity | Signs |
|---|---|---|
| I | **Preseptal cellulitis** | Lid edema, **no** proptosis / normal motility & vision |
| II | **Orbital (postseptal) cellulitis** | Proptosis, chemosis, ± ophthalmoplegia |
| III | **Subperiosteal abscess** | Displaced globe; medial most common (ethmoid) |
| IV | **Orbital abscess** | Severe proptosis, **↓ vision**, ophthalmoplegia |
| V | **Cavernous sinus thrombosis** | Bilateral signs, CN **III**/**IV**/**V**/**VI** palsies, picket-fence fever |

**PEARL** Most orbital complications arise from **ethmoid sinusitis** (thin lamina papyracea). **Best test = contrast CT orbits/sinuses.** Preseptal (I) → IV antibiotics; abscess (III–IV) or vision threat → **surgical drainage**. Spread to the **contralateral** eye = cavernous sinus thrombosis until proven otherwise.

## 7. Juvenile Nasopharyngeal Angiofibroma (JNA)

- **Adolescent males**; benign but **locally aggressive**, highly vascular. Classic: recurrent **epistaxis** + **unilateral nasal obstruction**.
- **Origin:** sphenopalatine foramen / **pterygopalatine fossa**.
- **Blood supply:** **internal maxillary artery** (ECA); large tumors may recruit ICA.
- **Imaging:** CT + MRI with contrast (avid enhancement). **Holman-Miller / antral bow sign** = anterior bowing of posterior maxillary wall. **Do NOT biopsy** (hemorrhage).
- **Staging: Radkowski** (or Fisch/UPMC) — based on extension & skull-base/ICA involvement.
- **Management: pre-op embolization (24–48 h before) → surgical excision** (endoscopic for most).

**RED FLAG** Never biopsy a suspected JNA in clinic — image first. A "nasal polyp" in a teenage boy that bleeds is JNA until proven otherwise.

## 8. Facial Fractures

### Zygomaticomaxillary complex (ZMC) — "quadripod"

- Four articulations/sutures: **zygomaticofrontal, zygomaticomaxillary, zygomaticotemporal, zygomaticosphenoid.**
- Signs: cheek flattening, trismus, **infraorbital nerve (V2) paresthesia**, lateral subconjunctival hemorrhage, diplopia.

### Le Fort fractures — all involve the pterygoid plates

| Type | Pattern | Hallmark |
|---|---|---|
| **I (Guérin)** | Horizontal, above teeth | "Floating palate" |
| **II** | Pyramidal | **Crosses infraorbital rim**; "floating maxilla" |
| **III** | Craniofacial disjunction | Midface separated from skull base; through zygomatic arch + nasofrontal suture |

**PIMP** "What's common to all Le Fort fractures?" → **pterygoid plate involvement.** "Which Le Fort involves the inferior orbital rim?" → **II.** Watch for **CSF rhinorrhea** in Le Fort II/III (cribriform).

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## 9. Oral Cavity, Neck Dissection & Flaps

### Oral cavity subsites

Lips (vermilion), buccal mucosa, upper/lower alveolar ridges, **retromolar trigone**, hard palate, floor of mouth, anterior ⅔ tongue.

### Neck nodal levels (I–VI)

| Level | Region |
|---|---|
| I (A/B) | Submental / submandibular |
| II–IV | Upper / mid / lower jugular (deep cervical) chain |
| V (A/B) | Posterior triangle / supraclavicular |
| VI | Central (pre/paratracheal, prelaryngeal/**Delphian**) |

- **Supraomohyoid** = I–III · **Lateral** = II–IV · **Posterolateral** = II–V · **Central** = VI.
- **Radical ND (RND):** levels I–V + **SCM, IJV, CN XI**. **Modified RND** spares ≥1 of those three. **Selective ND** spares nodal levels.

### Reconstructive flap blood supply

| Flap | Pedicle |
|---|---|
| Submental island | Facial → **submental artery** |
| **Pectoralis major (PMMF)** | **Thoracoacromial (pectoral branch)** |
| Deltopectoral (Bakamjian) | Perforators of **internal mammary** (1st–4th) |
| Radial forearm (free) | Radial artery |

## 10. Tongue Anatomy

### Muscles

- **Extrinsic (position) — all CN XII except palatoglossus:** genioglossus **protrudes**, hyoglossus **depresses**, styloglossus **retracts**; **palatoglossus elevates — CN X (exception).**
- **Intrinsic (shape):** superior & inferior longitudinal, transverse, vertical.

### Papillae

| Type | Taste? | Note |
|---|---|---|
| **Filiform** | No | Most numerous; mechanical only |
| Fungiform | Yes | Tip & sides |
| **Circumvallate** | Yes | Largest; V-shape anterior to sulcus terminalis |
| Foliate | Yes | Posterolateral |

**PEARL** Anterior ⅔ general sensation = **lingual (V3)**; taste = **chorda tympani (VII)**. Posterior ⅓ sensation + taste = **CN IX**. Motor = **CN XII** (all except palatoglossus = X).

## 11. Salivary Glands & Facial Nerve

| Gland | Duct | Secretion | Secretomotor |
|---|---|---|---|
| Parotid | **Stensen** | Serous | **CN IX** (otic ganglion) |
| Submandibular | **Wharton** | Mixed (mostly serous) | **CN VII** (chorda tympani → submandibular ganglion) |
| Sublingual | Rivinus (+ Bartholin) | Mixed (mostly mucous) | **CN VII** |

### Facial nerve (CN VII) — intratemporal segments

- Meatal → **labyrinthine (shortest & narrowest — most vulnerable)** → tympanic → mastoid.
- **Geniculate ganglion → greater petrosal nerve → lacrimal gland** (tearing).
- Branches: greater petrosal, **nerve to stapedius**, **chorda tympani** (taste ant ⅔ + submandibular/sublingual secretion).
- Extracranial: splits at **pes anserinus** → temporofacial & cervicofacial → 5 branches (Temporal, Zygomatic, Buccal, Marginal mandibular, Cervical).

**PIMP** "Shortest facial nerve segment / commonest bottleneck in Bell's palsy?" → **labyrinthine.** "Parotid vs submandibular secretomotor nerve?" → **IX vs VII.** Post-thyroidectomy tetany (Chvostek/ Trousseau) = **hypocalcemia** from parathyroid injury.

## 12. Tonsils & Adenoids

- **Waldeyer's ring:** pharyngeal (adenoid) + paired tubal + paired palatine + lingual tonsils.
- Palatine tonsil blood supply mainly from the **tonsillar branch of the facial artery**; main bleeding risk in tonsillectomy is the **lower pole**.

### Paradise criteria (recurrent tonsillitis → tonsillectomy)

- **≥7 in 1 yr, or ≥5/yr × 2 yr, or ≥3/yr × 3 yr**, with documented sore throat + one of: temp >38.3°C, cervical adenopathy, tonsillar exudate, or +GAS culture.

**RED FLAG** **Post-tonsillectomy hemorrhage:** primary (<24 h) vs secondary (~POD 5–10, eschar separation). Any bleeding → OR/control; don't wait.

## 13. Thyroid — Anatomy, TI-RADS, Bethesda

### Anatomy pearls

- Blood supply: **superior thyroid artery** (ECA) & **inferior thyroid artery** (thyrocervical trunk); rare **ima** artery.
- **RLN runs in the tracheoesophageal groove; closely related to the inferior thyroid artery & ligament of Berry** (commonest injury site). External branch of superior laryngeal nerve ↔ superior thyroid vessels (voice pitch).

### ACR TI-RADS (US risk)

| Feature | Points |
|---|---|
| Composition | cystic/spongiform 0 · mixed 1 · solid 2 |
| Echogenicity | an/hyper 1 · iso 1 · hypo 2 · very hypo 3 |
| Shape | wider-than-tall 0 · **taller-than-wide 3** |
| Margin | smooth/ill-defined 0 · lobulated/irregular 2 · ETE 3 |
| Echogenic foci | none/comet 0 · macrocalc 1 · peripheral 2 · **punctate 3** |

| Sum | Level | FNA if | Follow if |
|---|---|---|---|
| 0 | TR1 benign | — | — |
| 2 | TR2 | — | — |
| 3 | TR3 | ≥2.5 cm | ≥1.5 cm |
| 4–6 | TR4 | ≥1.5 cm | ≥1.0 cm |
| ≥7 | **TR5** | **≥1.0 cm** | ≥0.5 cm |

### Bethesda system (FNA)

| Cat | Diagnosis | Malignancy risk* | Management |
|---|---|---|---|
| I | Nondiagnostic | 5–10% | Repeat FNA (US-guided) |
| II | Benign | 0–3% | Follow-up |
| III | AUS/FLUS | ~10–30% | Repeat FNA / molecular |
| IV | Follicular neoplasm | ~25–40% | **Lobectomy** (Dx: need histology for capsular/ vascular invasion) |
| V | Suspicious for malignancy | ~50–75% | Lobectomy / near-total |
| VI | Malignant | 97–99% | Thyroidectomy |

*2017 Bethesda ranges (higher end if NIFTP counted as malignant). Catherine's notes use the older lower ranges — cite 2017 on exams.

PEARL **FNA is the best initial test** for a thyroid nodule (after TSH + US). **Follicular vs Hürthle carcinoma cannot be diagnosed on FNA** — needs histology showing capsular/vascular invasion (that's why category IV → diagnostic lobectomy).

## 14. Thyroid Cancers

| Type | Cell / marker | Spread | High-yield |
|---|---|---|---|
| **Papillary** (most common, ~80%) | Follicular epithelium | Lymphatic | **Psammoma bodies, Orphan-Annie nuclei, nuclear grooves;** best prognosis; risk: radiation, **FAP**/ **Gardner** |
| Follicular (2nd) | Follicular epithelium | **Hematogenous** (bone, lung) | Dx by **capsular/vascular invasion** |
| Hürthle cell | Oncocytic | Lymph/vascular | Follicular variant; less RAI-avid |
| **Medullary** | Parafollicular **C-cells** → **calcitonin** (+ CEA) | Lymph/blood | **Amyloid stroma; MEN2A/2B, RET** ; ~75% sporadic |
| Anaplastic | Undifferentiated | Aggressive local | Elderly; giant/spindle "bizarre" cells; poor prognosis |
| Primary lymphoma | Non-Hodgkin B-cell | — | Associated with **Hashimoto's** |

**PIMP** "Most common thyroid cancer / best prognosis?" **Papillary.** "Spreads hematogenously?" **Follicular.** "Calcitonin + amyloid + MEN2?" **Medullary** (screen RET, check pheo before surgery). "Rapidly enlarging neck mass in the elderly?" **Anaplastic.** "Cancer arising in Hashimoto's?" **Primary thyroid lymphoma.**

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**Scope note:** Reviewer reorganized from Catherine's notes, corrected and expanded to be Cummings-consistent and exam-ready. Numbers cross-checked against ACR TI-RADS, 2017 Bethesda System, AAO-HNS tonsillectomy guidance, and standard references. Where an edition-specific value matters (e.g., Bethesda risk %), defer to the current Cummings edition.